Carlos de Anda

Guadalajara, Mexico | carlosfelipedeanda@gmail.com | LinkedIn | GitHub

Computer Science student (GPA 9.65/10) with hands-on experience in enterprise networking, full-stack development, and IoT/computer vision. 1st place at Expo Ingenierías, 3rd place nationwide in robotics, and Vice President of FETEC (5,000+ students).

Education

Tecnológico de Monterrey (ITESM) Aug 2024 – June 2028 (Expected)
B.S. in Computer Science and Technology (ITC) — ***Cumulative GPA: 9.65 / 10.0*** *Guadalajara, Mexico*

Technical Skills

Languages: Python, C++, TypeScript, JavaScript, SQL, HTML/CSS, MATLAB, R
Frameworks & Libraries: React, Node.js, Express, OpenCV, Tailwind CSS, Vite
Networking & Tools: Cisco IOS, Packet Tracer, IPv4/VLSM, VLANs, Git, GitHub, Linux/Bash, MongoDB, MySQL
Spoken Languages: Spanish (Native), English (C1 Advanced), German (B1 Intermediate)

Technical & Engineering Projects

Smart Mirror System (1st Place at Expo Ingenierías) | *Python, OpenCV, React, Node.js, IoT* June 2026
- Engineered a full-stack smart mirror integrating computer vision, IoT devices, and embedded logic, winning 1st place in the Physical Prototype category at Tecnológico de Monterrey.
- Implemented real-time computer vision and facial recognition pipelines using Python, OpenCV, and `face_recognition` on an edge-computing Raspberry Pi central hub.
- Built IoT architecture connecting NodeMCU for PIR/LDR sensor telemetry and PWM-controlled lighting, integrated with a modular Node.js/Express backend, MongoDB Atlas, and MySQL.
- Developed an interactive dashboard UI using React, Vite, Tailwind CSS, and TypeScript with live API integrations. **[GitHub]**

Enterprise Campus Network Architecture | *Cisco IOS, IPv4/VLSM, Catalyst Switches, Wi-Fi 6* March 2026
- Designed end-to-end enterprise network architecture for a 6-story residential building supporting over 3,200 concurrent users and IoT devices.
- Designed the IP addressing scheme via VLSM (`172.24.0.0/18`) and structured 13 segmented VLANs for Wi-Fi, IoT, security, BMS, and VoIP services.
- Selected hardware including Cisco Catalyst C8300 Core Router, C9500 Core Switch, C9300 Access Switches (UPOE/Multigigabit), and 150+ indoor/outdoor Wi-Fi 6 APs.
- Estimated 1.5 Gbps WAN bandwidth requirements via oversubscription models and prepared a $338k+ USD financial proposal.
- Validated the network topology through a full-scale Cisco Packet Tracer simulation and deployed a physical Proof of Concept configuring trunking, DHCP, SSH, and routing.

Streaming Platform Engine Simulator | *C++, Object-Oriented Programming (OOP)* June 2025
- Designed and implemented a modular C++ streaming service simulation managing 50+ dynamic video objects.
- Applied OOP principles (inheritance, polymorphism, encapsulation) to build efficient rating, filtering, and content management algorithms.

Leadership & Engineering Achievements

Vice President — FETEC Executive Committee 2026 – Present
Tecnológico de Monterrey (Campus Guadalajara) *Guadalajara, Mexico*
- Co-led the winning executive campaign (Planilla NOVA) for the 2026–2027 student government term.
- Direct campus-wide student federation initiatives, managing strategic planning, budget execution, and representation for 5,000+ students.

Tec Ambassador (12th Generation) 2026 – Present
Institutional Representative | Tecnológico de Monterrey *Guadalajara, Mexico*
- Represent Campus Guadalajara in high-level institutional events, guiding prospective engineering students and visiting delegations.

EGS Global Leadership Program 2025 – 2026
Golden Dandelion Certification (Top 10% Performance) *Guadalajara, Mexico*
- Selected for the competitive 5th generation program; awarded Don Eugenio's Circle distinction for leadership and social impact.

TMR Lunar Rover Competition — Robotec Team May 2025
3rd Place Nationwide | Tecnológico de Monterrey *Mexico City, Mexico*
- Designed and programmed sub-systems for a lunar rover prototype, securing 3rd place nationally at the Mexican Robotics Tournament.

FIRST Robotics Competition — Daedalus 7102 2023 – 2024
Team Member | Houston World Championship Qualifier *Monterrey, Mexico*
- Contributed to robot mechanism assembly and strategic drive team operations, qualifying for the FRC World Championship in Houston.